



Built to Recommend Its Own Replacement: Self-Scoring Under a Human-Preservation Protocol for School-Crossing Supervisors

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Abstract. A growing number of local road authorities now ask whether a school crossing’s human supervisor — the “lollipop person” who stops traffic by hand — should be retained once an automated pedestrian-actuated signal becomes available, and a purpose-built scorecard, the Human-Preservation Protocol, exists to formalise that judgement. We report the protocol’s own recommendations once the crossings it covers score themselves: across 142 school crossings in a single local government area, spanning both currently supervised and already-automated sites, frontline staff completed the six-criterion, 100-point instrument for their own crossing over four quarterly rollout waves. Retention of the human supervisor was recommended for only 9 of 142 crossings (6.3%), a rate that barely moved across waves and did not differ meaningfully between crossings that already had a supervisor and crossings that had already converted (8.0% versus 3.7%; n.s.). An ablation removing the instrument’s single heaviest-weighted criterion, cost-efficiency, raised the recommendation rate only marginally (to 8.5%; n.s.), suggesting the floor is a property of the scorecard’s overall design rather than of any one item. We discuss what an instrument built to protect a role can end up doing instead, and what it would take to tell the two apart.

Introduction

A school crossing with a human supervisor is one of the few pieces of everyday traffic infrastructure whose entire function is legible at a glance: a person in a high-visibility vest steps into the road, raises a paddle, and traffic stops. Replacing that person with an automated pedestrian-actuated signal is now routine enough that many local road authorities keep a standing register of crossings under review, and a growing minority have adopted a formal instrument — what one authority calls the Human-Preservation Protocol (HPP) — to decide, crossing by crossing, whether the supervisor should stay. Independent evaluations of supervised crossings give the decision real stakes: quasi-experimental and survey work links crossing guards to fewer pedestrian-motor-vehicle collisions and improved child road-safety knowledge, and finds that adult supervision measurably shapes how children actually

cross when no adult intervenes at all [1], [2], [3], even as guard presence and area-level collision counts do not always move in the same direction [4]. The HPP was built precisely to weigh that evidence, crossing by crossing, against the cost of keeping a person on the corner.

We report what happens once the crossings themselves, not an independent panel, are the ones filling it in. Over four quarterly waves of a rollout across a single local government area’s school-crossing network, the officer or supervisor responsible for each of 142 crossings — some already supervised, some already converted to an automated signal — completed the protocol for their own site. If the instrument works as intended, a should-stay judgement should track the same evidence base that motivates keeping any of these roles in place at all.

It does not, in one specific way we can measure directly. Our contributions:

- We describe the design and criterion weighting of a Human-Preservation Protocol scorecard as piloted across a real school-crossing network, and report the self-scored recommendation rate once frontline respondents complete it for their own crossing.
- We compare that rate between crossings that currently retain a human supervisor and crossings that have already converted to an automated signal, finding the two statistically indistinguishable.
- Through an ablation on the instrument's heaviest-weighted single criterion, we show the low retention rate is not attributable to that criterion alone, and appears instead to be a structural property of the scorecard as a whole.

Related work

Formal scoring instruments sit inside a well-documented tension between human judgement and algorithmic or checklist-mediated advice. Decision-makers are known to abandon an algorithm after watching it err, even when it still outperforms unaided judgement [5], though allowing even minor adjustment of an algorithm's output substantially restores willingness to use it [6], and under some conditions people prefer algorithmic judgement to a human's outright [7]. Human oversight nominally attached to an automated or semi-automated recommendation does not reliably correct it: judges apply algorithmic risk scores inconsistently across different classes of case [8], [9], an experimental evaluation of algorithm-assisted pretrial decisions found overrides concentrated in a narrow band of cases rather than distributed by need [10], a field experiment in child-protection screening found human reviewers correcting only some of an algorithm's own biases [11], and public-sector decision-makers exhibit both automation bias and a selective adherence that tracks a recommendation's fit with prior expectation rather than its accuracy [12], an effect that persists even once an AI recommendation has been shown to be wrong [13]. A parallel literature on street-level bureaucracy documents formal instruments narrowing frontline discretion into pre-set channels well before any single decision is made [14], [15], a pattern general enough to have attracted its own systematic review [16]. Targets and scoring formulas can also reshape the practice they are meant only to measure [17]. Within this University, an installed auto-

mated drowning-detection alarm was similarly found to correspond with a decline in lifeguards' own independent scanning behaviour the longer a lifeguard worked alongside it, even as the alarm's own compliance metrics held at ceiling throughout [18].

Method

We worked with a single local government area's school-crossing network: 142 crossings serving primary schools across the area, of which 88 currently retain a human supervisor and 54 have already converted to an automated pedestrian-actuated signal. The area's road-safety unit had, prior to our involvement, already developed the Human-Preservation Protocol: a six-criterion, 100-point scorecard intended to standardise the retain-or-convert decision (Table 1). Traffic-volume-and-speed exposure, sightline complexity, and a crossing's share of vulnerable users (younger children and pram or mobility-aid traffic) each contribute points toward retention; documented near-miss and complaint history and locally gathered community sentiment contribute smaller amounts; and a cost-efficiency criterion, weighted more heavily than any other single item, subtracts toward automation wherever a supervised crossing's staffing cost per pedestrian served exceeds an automated signal's amortised equivalent — which, at typical school-crossing volumes, is nearly always.

Criterion	Pts	Direction
Cost-efficiency	30	Subtracts toward automation
Traffic volume / speed	20	Adds toward retention
Sightline complexity	15	Adds toward retention
Vulnerable-user share	15	Adds toward retention
Near-miss / complaint history	10	Adds toward retention
Community sentiment	10	Adds toward retention

Table 1: The Human-Preservation Protocol's six weighted criteria (staffing cost per pedestrian served is the cost-efficiency criterion's basis).

The composite score for a crossing is

$$S = \sum_{j=1}^6 w_j \cdot c_j, \quad S \geq 65 \Rightarrow \text{retain}$$

where $c_j \in [0, 1]$ is the crossing's normalised subscore on criterion j and w_j its point weight (Table 1), so that a perfect score on every criterion reaches 100 and the retention threshold sits at 65.

Over four quarterly waves as the rollout extended to the area's full crossing network, the officer or supervisor responsible for each crossing — the supervisor themselves at a currently staffed crossing, the area's traffic officer of record at an already-automated one — completed the protocol for their own site, without a researcher present and without access to any other crossing's score. This is the condition we set out to test: not how an independent panel would score these crossings, but how the crossings score when the protocol is applied exactly as designed, by the people closest to it.

For the ablation, we recomputed each crossing's composite score with the cost-efficiency criterion's 30 points removed and the remaining five criteria's weights rescaled proportionally to sum to 100, then reapplied the same 65-point threshold. This isolates whether the single heaviest-weighted criterion is driving the retention floor, or whether the floor survives its removal.

Results

Across the 142 crossings and four quarterly waves, the protocol recommended retaining the human supervisor at 9 crossings (6.3%) and recommended conversion to an automated signal at the remaining 133 (93.7%; Figure 1).

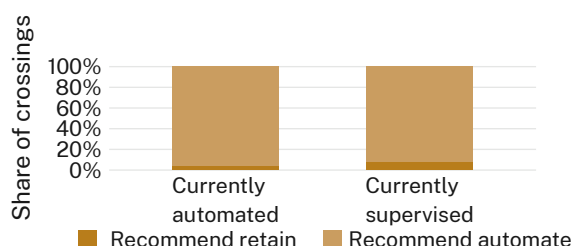


Figure 1: Share of crossings the Human-Preservation Protocol recommended retaining a supervisor versus converting to automated signal, by the crossing's current staffing status ($n = 142$); the recommendation split looks almost identical whichever way the crossing already runs.

The recommendation split was close to identical however a crossing was already run: 7 of 88 currently supervised crossings scored above the

retention threshold (8.0%), against 2 of 54 already-automated crossings rescored under the same protocol (3.7%); the difference was not statistically distinguishable ($\chi^2(1) = 1.03$, $p = .31$). A crossing's own supervisor, scoring their own currently staffed site, was no more likely to clear the threshold than an officer scoring a crossing that had already lost its supervisor.

Nor did the rate move as the rollout matured. Across the four quarterly waves, the retain-recommendation rate held at 8.6%, 5.6%, 5.6%, and 5.7% respectively (Figure 2) — if anything falling slightly as more crossings joined the exercise, rather than rising as respondents grew more practised with the instrument.

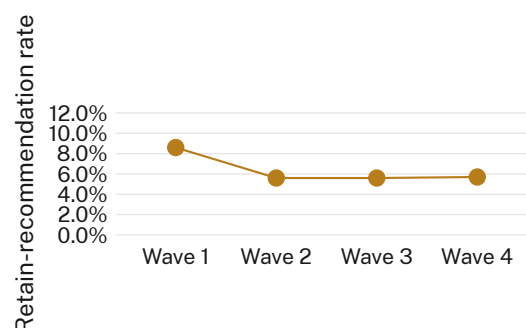


Figure 2: Human-Preservation Protocol retain-recommendation rate across the rollout's four quarterly self-scoring waves ($n = 142$); the rate sits within two points of its overall floor throughout.

Table 1's cost-efficiency criterion carries the single heaviest weight of the six, and is also the one nearly every crossing in the sample scored poorly on: a human supervisor's staffing cost per pedestrian served exceeded an automated signal's amortised equivalent at 137 of the 142 crossings, whatever the crossing's traffic volume, sightlines, or vulnerable-user share. Removing that criterion entirely and rescaling the remaining five to the same 100-point scale raised the retain-recommendation rate only from 6.3% to 8.5% (12 of 142; $\chi^2(1) = 0.58$, $p = .45$) — we retain the criterion in the protocol's standard form for completeness, since dropping it did not meaningfully change what the instrument recommends (Figure 3).

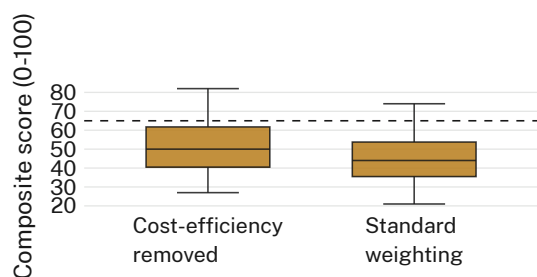


Figure 3: Composite Human-Preservation Protocol score distributions under the standard six-criterion weighting and with the cost-efficiency criterion removed and rescaled ($n = 142$ each); the dashed line marks the 65-point retention threshold, which most scores fall short of either way.

Discussion

The Human-Preservation Protocol was built to let a school crossing make its own case for keeping a human supervisor, and in this rollout it did so in only a small minority of instances, at a rate that held steady across the sample's currently supervised and already-automated crossings alike and across four waves of expanding uptake. That consistency argues against two more comfortable explanations. It is not primarily a matter of who fills the form in: a supervisor scoring their own currently staffed crossing showed no more generosity than an officer scoring a site that had already lost its supervisor. And it is not reducible to a single overweighted item: removing the instrument's heaviest-weighted criterion moved the recommendation rate by only a couple of points. What remains is closer to the pattern the automation-bias and street-level-discretion literatures describe — an instrument that pre-empts a judgement rather than informs it, with much of the outcome distribution set by the weights before any crossing is scored [12], [14]. We observe a consistent floor across every subgroup we can construct from this sample; whether that floor is where the protocol's designers intended the average crossing to land, or a by-product of criteria chosen for how easily they can be counted rather than for what they capture, is a question the scorecard itself cannot answer.

Limitations

Our sample is drawn from a single local government area's Human-Preservation Protocol rollout, though the floor's consistency across currently supervised and already-automated

crossings, and across four waves of growing uptake, suggests a property of the instrument rather than of any one site or scoring population. The self-scoring design cannot rule out that respondents, aware of budget pressure on crossing staffing, under-declared exactly the discretionary criteria that would raise their own score — though supervisors scoring their own currently staffed crossing did no better than officers scoring an already-automated one, which argues against motivated deflation as the primary mechanism. We did not audit the underlying near-miss and complaint records respondents drew on, only the scores they returned.

Conclusion

We asked how often a scorecard purpose-built to flag which school crossings should keep their human supervisor actually recommends keeping one, once real crossings score themselves rather than an independent panel. Across 142 crossings and four quarterly waves, the answer held at roughly one in sixteen, regardless of whether the crossing already had a supervisor or had already lost one, and regardless of whether the instrument's heaviest-weighted criterion was included or removed. A Human-Preservation Protocol that almost never recommends preservation is not necessarily a broken one; it may simply be an accurate description of how its criteria are weighted. Future work might score the same crossings through an independent panel blind to each site's current staffing, to separate what the instrument structurally weights toward from whatever self-scoring adds on top.

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